

Object-oriented Programming 2 KU

Introduction and Timeline

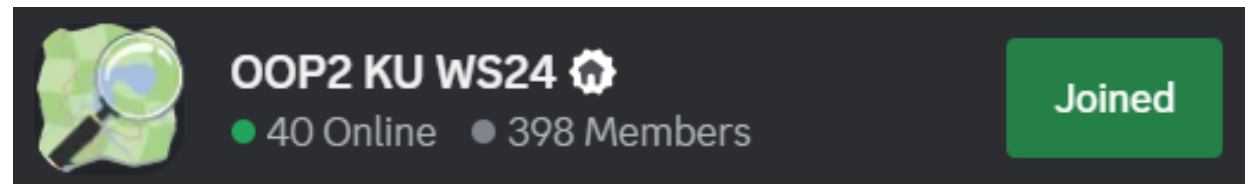
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Institute of Computer Graphics and Knowledge Visualization (CGV)

Resources

Resources

- [Teach Center](#):
 - Slides and code snippets will be uploaded there
 - Same TC course for VO and KU
- [Discord](#):
 - Weekly question hours (if there is high demand!)
 - Main communication over Discord (Announcements and FAQ)
 - Read and accept the server rules!



Resources

■ E-Mail

- For personal questions **only**, you can also write an E-Mail to:
- oop2ku-2024@cgv.tugraz.at

■ Repositories (!!)

- Sign up [here](#) to get the repositories today (TUG Login)

CGV Student GitLab

Computer Graphics and Knowledge Visualization

Submission system for courses (EEGS, GDI, SVU, OOP2, CAGD, ...)

- In case of login troubles, ask yourself if you are student or staff
- If you still have login troubles contact [Wolfgang Scheicher](#)
- The [Newsgroups](#) for [FGP](#), [GDI](#) and [SVU](#) are a good place for discussing the usage of this system



Staff

Nutzername

Passwort

☐ Login merken

Anmelden

oder melde dich an mit

TUGRAZ online (Students)

☐ Login merken

KU Timeline

KU Timeline

- Assignment 0:
 - 30 Points (check out the README)
 - Needs to be done individually!
 - At least 60% (18 points) of the points needed to continue the course!
Otherwise: De-registration from KU without a grade
 - Deadline: **24.10.2024 23:59**
 - NO second chance deadline! Strict deadline! (MRC at 23:59 will be used)
 - Make sure you submit it to the correct branch! (See README)
- After the A0 deadline, team registration will open in TC
 - Sign up in teams of 3 for the group assignments!
 - **After registering for a group in Teach Center, you will be graded!**

KU Timeline

- Assignment 1:
 - 36 Points
 - Groups of 3 (ONLY in exceptions with less participants!)
 - **After registering for a group in Teach Center, you will be graded!**
 - Make sure everyone participates! We will check! (git statistics, AG, ...)
 - In case of abnormal group participation: Contact us!
 - Soft Deadline: **19.12.2024 23:59**
 - **Test System will only test your A1 until the Soft Deadline!**
 - Hard Deadline together with A2

KU Timeline

- Assignment 2:
 - 34 Points
 - Same groups of 3 as for assignment 1 (Switching groups is **not** possible)
 - Builds on A1 framework and implementation
 - Hard Deadline together with Assignment 1: **16.01.2025 23:59**
- Bonus Assignment: (Independent Framework, 10 Points)
 - Same groups of 3 as for A1/A2
 - Hard Deadline with A2
 - Counts towards a positive grade
 - **(All bonus points except bonus points from the evaluation count towards a positive grade)**

KU Timeline

- Assignment Reviews:
 - In person in the week(s) after the hard deadline for A1/2
 - Will be announced in time!
 - Mandatory for every team member
 - All team members need to know everything about the entire submission
 - Questions about the code, programming tasks, comprehension questions
 - Questions about ALL implemented features and assignments (A0, Bonus)
 - NO second chance if negative due to assignment review
 - Plagiarism Talks will be held after assignment reviews

KU – Grading

```
public static Grade getGrade(float ass0, float ass1, float ass2, float bonus) {  
  
    if (ass0 < 18) {  
        System.exit(0); // Drop out of the course without getting a grade - You do not lose an attempt  
    }  
  
    float total = ass0 + ass1 + ass2 + bonus;  
  
    if (total >= 55) {  
        if (total >= 91) {  
            return Grade.SEHR_GUT; // 1  
        }  
        if (total >= 79) {  
            return Grade.GUT; // 2  
        }  
        if (total >= 67) {  
            return Grade.BEFRIEDIGEND; // 3  
        }  
        return Grade.GENUEGENDE; // 4  
    }  
    return Grade.NICHT_GENUEGENDE; // 5  
}
```

Tutorials and Q&As

Tutorials and Q&As

- KU Sessions:
 - Every Thursday in the lecture hall i13 (As stated in TUG-Online)
 - Mostly used for tutorials but also for questions about the assignments!
 - Might NOT be recorded
- Question Hours in Discord
 - Not recorded
 - Only if there is (high!) demand
 - Otherwise: Ask in the forum channels! Use the template for questions!



Tutorials and Q&As



- Discord Support:
 - Ask questions in the corresponding assignment channels
 - Stick to the template when asking questions!
 - Use the provided tags when creating a post
 - Questions that violate the template will be ignored/deleted
 - NO information about private test cases!
 - Read before posting a new question; it might already have been answered!

Assignment Overview



Assignment Overview

- Goal: apply OOP principles to a practical application:
 - Usage of modern libraries (Spring Boot, gRPC, (LWJGL for Bonus), ...)
 - Utilize modern Java principles (Streams, Efficient Data Structures, ...)
- Context: Geographic Information System (GIS) as a RESTful API
 - Your task: Build a system that can provide geographic information through a standardized API from an XML resource on the backend
 - Three assignments focusing on different aspects of the system:
 - A0: Basic objects, parameter parsing, error handling
 - A1: API implementation, networking, drawing, parsing
 - A2: Algorithms in Java, libraries in-depth
 - Bonus: GPU rendering with the help of [LWJGL](#)

OOP2 Mapping Demo



Data Loading

Amenities

Type:

id	name	type	geom
66196	Landesgericht für Zivilrechtssachen Graz	courthouse	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.4361, 47.06753, ...
67291	Technische Universität	university	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.44984, 47.0688, ...
1321222	Landesgericht für Strafsachen Graz	courthouse	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.445913, 47.060, ...
1809878	Kindergarten Josef-Huber-Gasse	kindergarten	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.421861, 47.066, ...
3366181	TU Graz - Physik Institute	university	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.45315, 47.0646, ...
4546039	Berufsschule St. Peter	school	["type": "Polygon", "coordinates": [[[[[15.462835, 47.060562], [15.463604, 47.060345], [15.463902, 47.060471], ...
4576210	Technische Universität Graz	university	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.451035, 47.069, ...
5156343		parking	["type": "Polygon", "coordinates": [[[[[15.449213, 47.053955], [15.44934, 47.054008], [15.44953, 47.054092], [1, ...
9526724	Congress Graz	events_centre	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.437218, 47.069, ...
9963571	Volksschule Schönaau	school	["type": "GeometryCollection", "geometries": [{"type": "MultiPolygon", "coordinates": [[[[[15.440792, 47.052, ...

Assignment 0

- Environmental variables (5P):
 - Make sure to properly parse them! Hint: `System.getenv().getOrDefault()`
 - This task involves handling environment variables for both the middleware and the backend services. You must ensure these variables are loaded, parsed, stored, and used appropriately, with proper logging. If any variables are missing or invalid, you should revert them to their default values.
 - Make sure to use the correct defines for middleware and backend!

Assignment 0

- Mock responses (18 Points):
 - Return data in the correct format - no actual data required for now (Look at the RESPONSES.md in the framework)
 - Do NOT just return the response as a hard-coded string! → **ZERO POINTS**
 - For A0 you will implement only the middleware part of these requests because the connection to the dataset is not yet set up. You will need to implement the request for 5 amenities and 5 roads
 - Just get started with Spring Boot and the REST framework

Assignment 0

- Mock responses (18 Points):

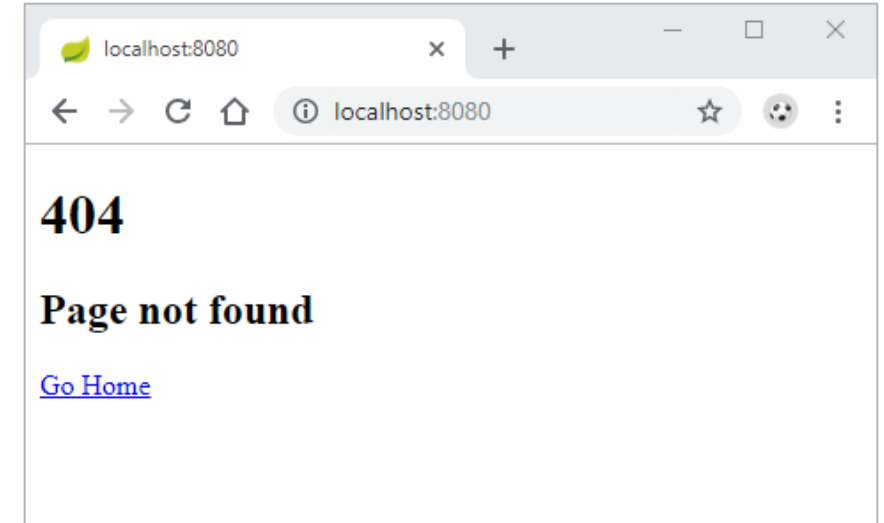
Amenity by ID

GET http://localhost:8010/amenities/26899640

```
{
  "name": "Polizeiinspektion Schmiedgasse",
  "id": 26899640,
  "geom": {
    "type": "Point",
    "crs": {
      "type": "name",
      "properties": {
        "name": "EPSG:0"
      }
    },
    "coordinates": [
      15.4393231,
      47.0686148
    ]
  },
  "tags": {
    "wheelchair": "yes",
    "addr:housenumber": "26",
    "phone": "+43 59133 6593 100",
    "amenity": "police",
    "addr:country": "AT",
    "name": "Polizeiinspektion Schmiedgasse",
    "addr:street": "Schmiedgasse",
    "addr:postcode": "8010",
    "addr:city": "Graz"
  },
  "type": "police"
}
```

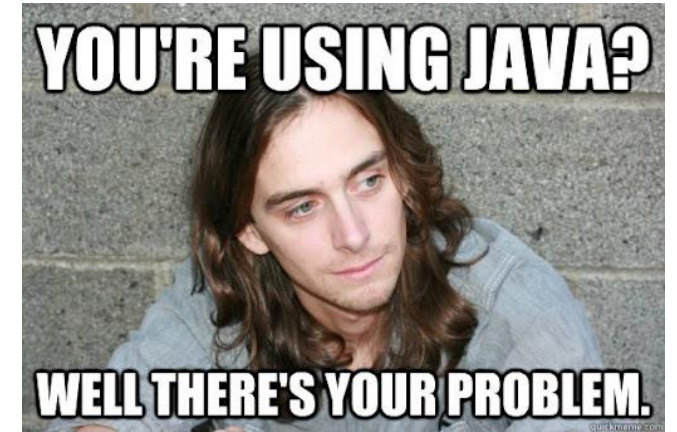
Assignment 0

- Error Handling (7 Points):
 - Wide variety of errors to consider (Status 400 and 404)
 - Important: Choose the correct return code!
 - If an error occurs, you should return a simple error response in the Spring Boot service as JSON, following the guidelines outlined in RESPONSES.md
 - The JSON response should indicate the error encountered, and most importantly the appropriate status code should be set.



Assignment 0 – Early Hints

- Setup:
 - Use Java 17 (SDK 17 in IntelliJ)
 - Check the [Java Cheatsheet](#) and sample [Java file](#) in TC if you are new to Java
 - Check the Spring Boot Tutorial from last year ([spring-boot.zip](#)) (Recording found on [TUbe](#))
 - Online resources that are helpful:
 - Will be posted in Discord under [#tutorials](#)

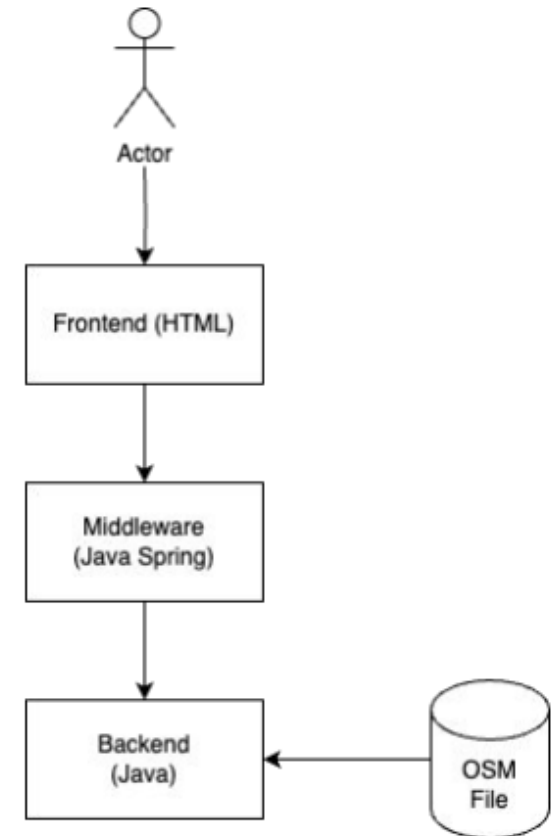


Assignment 0 – Early Hints

- Mock responses
 - Use [JSONDiff](#) for testing the JSON responses for diffs (Order does not matter)
 - Use [Insomnia](#) (open source, testing REST requests)
 - Alternative: [Postman](#)
- Environmental arguments
 - Make sure to properly parse them! Hint: `System.getenv().getOrDefault()`
 - Look at the [sample code in TC](#) and [TUBE](#) for Spring Boot
 - Handle invalid ports! Do not expect integers as ports, and do not throw errors/stop -defaults!
 - Please use the Logger functions! (A0: two must be used)
- Error Handling
 - Read through [this article](#)

Assignment 0 – Early Hints

- Setup:
 - Use IntelliJ/Any IDE to Debug/Build your solution, Docker to only verify
 - [IntelliJ](#) (Free student license)
 - Docker build might throw weird errors. Try to clean cache
 - The HTML file frontend is already complete, **no need to touch it!**
- Where to start:
 - Go through the README and the linked Tutorials in Discord
 - Look at the [sample code in TC](#) and [TUBE](#) for Spring Boot
 - Last year's recordings in [Tube](#)



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